

WILLIAM JACKSON

Senior Software Engineer

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PROFESSIONAL SUMMARY

DevOps and cloud infrastructure engineer with strong experience building scalable delivery platforms and production automation across modern SaaS and distributed systems environments. Over years of working in high-traffic software and platform engineering teams, expertise developed around **DevOps, AWS Cloud, Docker, Kubernetes, CI/CD automation, and Infrastructure as Code** while supporting backend engineering teams delivering microservices and API platforms. Extensive hands-on work modernizing legacy deployment systems into cloud-native environments using **Terraform, GitHub Actions**, and container orchestration while improving reliability, monitoring, and release velocity across engineering organizations. Proven ability to troubleshoot complex deployment failures, resolve infrastructure incidents, and optimize system observability using modern monitoring stacks, enabling highly resilient production environments and faster engineering delivery cycles.

SKILLS

- **DevOps & Infrastructure:** Docker (18-24), Kubernetes (1.18-1.29), Terraform (0.12-1.x), Helm, Linux, Bash, Infrastructure as Code, Microservices Deployment
- **Cloud Platforms:** AWS (EC2, EKS, ECS, Lambda, API Gateway, S3, RDS, DynamoDB, CloudWatch), Azure exposure, cloud networking and security architecture
- **CI/CD & Automation:** GitHub Actions, Jenkins, GitLab CI, Argo CD, automated pipelines, build orchestration, artifact management
- **Monitoring & Observability:** Prometheus, Grafana, Datadog, ELK Stack, CloudWatch, distributed logging and alerting systems
- **Programming & Scripting:** Python, Node.js, Shell scripting, infrastructure tooling development and automation utilities
- **Data & Messaging:** PostgreSQL, MySQL, Redis, Kafka, RabbitMQ, distributed system integration

WORK HISTORY

Senior Software Engineer

February 2021 to December 2025

Viaro Networks Inc. - Delaware, United States

- Led transformation of legacy deployment workflows into a fully automated **CI/CD pipeline** using **GitHub Actions** and **Docker (19-24)** containers, eliminating manual deployment steps and significantly improving release reliability across multiple microservice platforms.
- Designed and implemented production-grade **Kubernetes (1.20-1.28)** clusters on **AWS EKS**, solving frequent service restart issues caused by resource misconfiguration by implementing optimized pod resource limits and autoscaling policies.
- Modernized infrastructure provisioning by migrating manually configured cloud environments into **Terraform (0.12-1.x)** infrastructure-as-code modules, enabling consistent environment recreation and drastically reducing configuration drift issues.
- Implemented centralized **logging and monitoring** using **Prometheus, Grafana**, and **ELK stack**, resolving difficult debugging challenges caused by fragmented log systems across distributed microservices.
- Optimized container build pipelines by introducing layered **Docker caching strategies**, fixing build failures triggered by dependency mismatches while improving pipeline build time significantly.
- Collaborated with backend teams developing **Node.js** and **Java microservices** to design deployment strategies that supported high-availability rollouts and blue-green release patterns across production clusters.
- Resolved recurring production incidents caused by database connection exhaustion by implementing connection pooling and infrastructure monitoring alerts integrated with **CloudWatch** and **Datadog**.
- Improved platform security by integrating automated container vulnerability scanning into **CI/CD pipelines**, preventing deployment of images containing high-risk dependencies.
- Led migration of legacy monolithic services into containerized **microservice architecture**, resolving several environment inconsistencies that previously caused deployment rollbacks.
- Implemented infrastructure cost optimization strategies by restructuring unused cloud resources and introducing automated shutdown schedules for non-production environments.
- Built automated disaster recovery infrastructure using **AWS snapshots and infrastructure replication**, ensuring platform reliability during regional service disruptions.
- Mentored engineers across development teams on **DevOps practices**, helping teams design deployment-ready services and maintain infrastructure observability standards.

Software Engineer
Avantia, Inc. - Ohio, United States

October 2018 to January 2021

- Engineered internal **deployment automation pipelines** using **Jenkins** integrated with **Docker (17-20)** environments to standardize application releases across multiple backend services.
- Implemented centralized **log aggregation** using **ELK stack**, solving debugging delays caused by application logs being distributed across dozens of isolated servers.
- Developed infrastructure monitoring dashboards with **Prometheus and Grafana**, enabling early detection of performance bottlenecks affecting API response latency.
- Migrated several legacy on-premise deployments into **AWS EC2 and containerized infrastructure**, resolving persistent configuration inconsistencies between staging and production environments.
- Diagnosed and fixed intermittent production outages caused by message queue overload in **RabbitMQ**, introducing queue monitoring and auto-scaling consumer workers.
- Built automated database backup pipelines for **PostgreSQL and MySQL**, preventing data loss risks and improving operational reliability for mission-critical services.
- Collaborated with backend teams deploying **Node.js APIs** and **Java services**, ensuring consistent infrastructure environments across development and production.
- Improved system reliability by implementing automated rollback procedures for failed deployments triggered by pipeline health checks.
- Designed secure secret management solutions using encrypted environment configurations integrated with deployment pipelines.
- Reduced infrastructure provisioning time by creating reusable cloud configuration templates for application environments.
- Documented operational playbooks for incident response and infrastructure troubleshooting across engineering teams.

Software Developer
ArnAmy, Inc. - Florida, United States

March 2016 to September 2018

- Developed backend automation utilities in **Python** and **Shell scripting** to streamline server provisioning and reduce repetitive infrastructure setup tasks.
- Implemented automated deployment scripts that eliminated manual file transfers and reduced deployment errors affecting application updates.
- Diagnosed performance bottlenecks caused by inefficient database queries in **MySQL**, collaborating with backend developers to optimize indexing and query execution.
- Introduced basic **Docker containerization** for development environments to standardize dependency management across engineering teams.
- Built internal monitoring scripts that tracked service uptime and alerted engineering teams when system health thresholds were exceeded.
- Collaborated with development teams building early microservice components and ensured infrastructure configuration aligned with application requirements.
- Resolved environment configuration issues that frequently caused build failures during staging deployments.
- Helped introduce version control workflows for infrastructure scripts using **Git**, improving collaboration across engineering teams.
- Contributed to backend services written in **Node.js**, assisting in debugging runtime errors affecting deployment stability.
- Improved internal documentation covering deployment procedures and system architecture diagrams.

Junior Software Developer
Centric Tech Inc. - New Jersey, United States

October 2014 to February 2016

- Assisted in maintaining backend services running on **Linux servers**, diagnosing deployment issues caused by inconsistent server configurations.
- Developed small automation scripts that helped engineering teams package application releases and prepare deployment artifacts.
- Supported senior engineers in maintaining database systems and monitoring application health across staging environments.
- Helped debug production incidents caused by configuration errors and assisted in restoring services quickly during outages.
- Implemented minor infrastructure improvements including log rotation scripts and backup verification processes.
- Collaborated with development teams to troubleshoot integration issues between application services and database systems.
- Assisted with deployment coordination for internal software releases and documentation of deployment processes.
- Helped maintain CI server environments and troubleshoot build failures affecting development pipelines.
- Improved operational documentation covering infrastructure troubleshooting and deployment workflows.

EDUCATION

Bachelor's degree in Computer Science

2010 to 2014

Stanford University Department of Computer Science

- Built strong foundations in **algorithms**, **databases**, and **software engineering**, applying coursework through practical team projects and production-style system design.

PROJECTS

Cloud-Native CI/CD Platform for Microservices

- Designed and implemented a **cloud-native DevOps platform** that automated build, test, and deployment pipelines for multiple microservices using **Docker (19-24)**, **Kubernetes (1.21-1.28)**, and **GitHub Actions**, solving frequent deployment failures caused by manual releases and inconsistent environments across development and production systems.
- Integrated **Terraform (0.13-1.x)** infrastructure modules and **Prometheus + Grafana monitoring**, enabling automated environment provisioning and real-time infrastructure observability while improving deployment reliability and reducing release time through fully automated **CI/CD pipelines** running on **AWS EKS**.